Determination of Theoretical Kalman Filter Performance for Atomic Clock Estimation Through Equivalent Linear Time-Invariant Systems

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Kalman filters (KFs) are widely used in atomic clock estimation. However, their theoretical performances are not strictly determined by traditional methods when the observation models and process (Q) and observation (R) noise variances are not consistent with actual observations. This article proposes a new method of studying the classic two-state KF in these cases. The core idea is that a KF in the steady state can be treated as equivalent to three linear time-invariant (LTI) systems. Then, the theoretical performance of each term in the KF estimates can be accurately and independently determined by using the superposition and homogeneity of the LTI systems, whose structures and coefficients are completely determined by the KF model and chosen Q and R values and independent of actual observations. An example is used in which three kinds of terms are involved in the observation, and three submethods are proposed. For the polynomial and periodic fluctuation terms, the analytical expressions of corresponding terms in the KF estimates are derived. For the noise term, the noise filtering performance is accurately determined in the frequency domain. By combining the new and traditional methods, the complete conclusions are obtained. The experimental results validate the theoretical conclusions. The research indicates that this method can effectively determine the theoretical performances of KFs when the KF model and Q and R values are not consistent with actual observations.

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I. INTRODUCTION

Atomic clocks are precision instruments that play important roles in timekeeping, satellite navigation, deep space exploration, and other electronic systems [1], [2], [3]. Kalman filters (KFs) are widely used to obtain the estimates of atomic clocks or time scales with respect to a reference time scale [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17]. The observation of an atomic clock or time scale relative to a reference time scale is input into the KF, and then the KF outputs are obtained, which are KF estimates, such as time and frequency estimates [3], [4]. Subsequently, these estimates are used to analyze clock performance, predict future values [5], [6], [7], correct (steer) clocks [7], [8], [9], [10], [11], [12], [13], detect anomalies [14], [15], [16], [17], and generate ensemble time scales [18].

In traditional methods, KFs are treated as optimal estimators in the time domain [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [19], [20], [21]. According to signal processing theory [22], [23], when the KF model and its process ($\boldsymbol{Q}$) and observation (R) noise variances are completely consistent with actual observations, the mean-squared errors of the estimates reach their minima.

The question then arises as to whether one can use a KF to estimate the states effectively or determine the theoretical performance when the KF model and $\boldsymbol{Q}$ and R values do not agree with actual observations. In practice, the KF model is not always consistent with actual observations. For example, classic two- [3], [4], [10], [11] and three-state KFs [5], [6] do not model periodic fluctuations. However, significant periodic fluctuations are commonplace in the observations of global navigation satellite system (GNSS) onboard atomic clocks [24], [25], [26] and the observations between two remote time scales obtained by two-way satellite time and frequency transfer [27]. In these cases, will the estimate performance deteriorate due to the unmodeled periodic fluctuations? In addition, a KF serves not only as an optimal estimator in the time domain, but also as a filter to smooth out noise [3], [4], [27]. Unfortunately, the effects of the $\boldsymbol{Q}$ and R values [19], [20], [28], [29] on the KF estimation and filtering performances have not been rigorously determined in theory.

To address this shortcoming, this article proposes a new method that adopts the idea of deriving the KF transfer functions in the previous literature by the authors [30], [31]. The classical two-state KF is used as the research object. The KF in the steady state is no longer treated as an optimal estimator in the time domain, but rather is taken to be equivalent to three linear time-invariant (LTI) systems through its transfer functions.

The LTI systems are fully determined by the KF model and its $\boldsymbol{Q}$ and R values, independent of actual observations. Because LTI systems satisfy superposition and homogeneity properties, each term in the KF estimates can be theoretically determined from the LTI system perspective,

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regardless of whether the KF model and $\boldsymbol{Q}$ and R values are consistent with actual observations.

An example is used in which the observation involves three kinds of terms: polynomial, periodic fluctuation, and noise terms, and is inconsistent with the KF model due to the introduced periodic fluctuation term. The first two are deterministic terms, and the last one is a stochastic term. Three submethods are proposed for the three terms to determine the theoretical performances of their corresponding terms in the KF estimates. Then, by combining the conclusions of the new and traditional methods, the complete conclusions for the theoretical KF performance are obtained.

The rest of this article is organized as follows. Section II describes the background related to the classic two-state KF, which constitutes the focus of this study. Section III discusses the core idea of the method, presents a derivation of the equivalent LTI systems, and provides an analysis of the derivation process. Section IV describes the actual observation model used in this study and discusses the three submethods for the three kinds of terms in the actual observations. Section V summarizes the conclusions by combining the new and traditional methods. In Section VI, a real measured data experiment is presented to validate the findings. Section VII provides a discussion of the future plans based on the study. Finally, Section VIII concludes this article.

II. CLASSIC TWO-STATE KF

This section describes the background of the classic two-state KF. Section II-A introduces the observation and process equations of the KF, as well as the observation $z(t_k)$ and states $x(t_k)$ and $y(t_k)$. Section II-B describes the instantaneous frequency, which is different from state $y(t_k)$. Section II-C addresses the approaches used to estimate the diffusion coefficient values, which determine the $\boldsymbol{Q}$ and R values. Section II-D describes the $\boldsymbol{Q}$ and R matrices. Finally, Section II-E summarizes the KF principle of atomic clock estimation.

A. KF Process and Observation Equations

The classic two-state clock model, which corresponds to the classic two-state KF, can be written in the form of a solution to stochastic differential equations [2], [32], [33], [34] as

$$\begin{cases} x(t) = x_0 + y_0 t + \frac{1}{2} d t^2 + \sigma_1 W_1(t) + \sigma_2 \int_0^t W_2(s) \mathrm{d}s \\ y(t) = y_0 + dt + \sigma_2 W_2(t) \end{cases} \tag{1}$$

where $x(t)$ represents the time state; $y(t)$ represents the frequency state at time t; x_0 and y_0 are the initial values of the time and frequency states, respectively; d represents the frequency drift; $W_1(t)$ and $W_2(t)$ are two independent one-dimensional standard Wiener processes; and σ_1 and σ_2 are the diffusion coefficients.

State $x(t)$ represents the instantaneous time of the atomic clock against a reference time scale. The observation of $x(t)$ appears as [3], [4], [32]

$$z(t_k) = x(t_k) + \sigma_0 \xi_0(t_k) = x_0 + y_0 t_k + \frac{1}{2} d t_k^2 + \sigma_1 W_1(t_k) + \sigma_2 \int_0^{t_k} W_2(s) \mathrm{d}s + \sigma_0 d \frac{W_0(t_k)}{dt} \tag{2}$$

where $z(t_k)$ represents the observation of $x(t_k)$; $\xi_0(t_k)$ is the introduced observation noise, which is assumed to be white phase modulation (WPM) and can be written as the derivative of a Weiner process $W_0(t_k)$ (i.e., $\xi_0(t_k) = \mathrm{d}W_0(t_k)/\mathrm{d}t$); and σ_0^2 represents the measurement noise variance.

In fact, the observation model described by (2) is the same as that in [32, Sec. V], where $\sigma_1 W_1(t)$ represents white frequency modulation (WFM), $\sigma_2 \int_0^t W_2(s)\mathrm{d}s$ represents random walk frequency modulation (RWFM), and $\sigma_0 \xi_0(t_k)$ represents WPM. The first two represent the clock noises, and the last one represents the observation noise.

The KF process equation uses the iterative form of (1) as [2], [3], [4], [32], [33], [34]

$$\boldsymbol{s}(t_{k+1}) = \begin{bmatrix} x(t_{k+1}) & y(t_{k+1}) \end{bmatrix}^T = \begin{cases} x(t_{k+1}) = x(t_k) + y(t_k) T + \frac{1}{2} d T^2 \\ + \sigma_1 \left[W_1(t_{k+1}) - W_1(t_k) \right] \\ + \sigma_2 \int_{t_k}^{t_{k+1}} [W_2(s) - W_2(t_k)] \mathrm{d}s \\ y(t_{k+1}) = y(t_k) + dT + \sigma_2 \left[W_2(t_{k+1}) - W_2(t_k) \right] \end{cases} \tag{3}$$

where $\boldsymbol{s}$ represents the two states in matrix form, t_k represents the time at the kth epoch, and $T = t_{k+1} - t_k$ is the sampling time for $k = 0, 1, 2 \ldots$.

To summarize, (2) and (3) are the observation and process equations of the KF, respectively.

B. Instantaneous Frequency

Note that state $y(t)$ does not represent the instantaneous frequency, which is different from the fact that state $x(t)$ represents the instantaneous time. The instantaneous frequency is represented by the derivative of $x(t)$, as follows:

$$\frac{dx(t)}{dt} = y_0 + dt + \sigma_1 \frac{dW_1(t)}{dt} + \sigma_2 W_2(t) \tag{4}$$

where $\sigma_1 \mathrm{d}W_1(t)/\mathrm{d}t$ represents WFM and $\sigma_2 W_2(t)$ represents RWFM in the instantaneous frequency.

Comparing (4) with (1), state $y(t)$ does not involve WFM, but rather only involves RWFM. Therefore, state $y(t)$ represents only part of the instantaneous frequency, which is generally called the "random walk" part of the instantaneous frequency [2], [32], [33], [34].

C. Estimating $(\sigma_0^2, \sigma_1^2, \sigma_2^2)$ Values

Before running the KF, the $\boldsymbol{Q}$ and R values must be determined by using the $(\sigma_0^2, \sigma_1^2, \sigma_2^2)$ values of the atomic clock. In practical applications, the $(\sigma_0^2, \sigma_1^2, \sigma_2^2)$ values can be estimated according to the slopes and intercepts of the Allan variance in the time domain and the power spectral densities (PSDs) in the frequency domain.

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1) *Time Domain:* The Allan variance is typically employed to characterize the frequency stability of an atomic clock in the time domain. If the atomic clock only involves WPM, WFM, and RWFM without other contributing terms, the Allan variance can be expressed as [4]

$$\sigma_y^2(\tau) = \frac{3\sigma_0^2}{\tau^2} + \frac{\sigma_1^2}{\tau} + \sigma_2^2\frac{\tau}{3} \tag{5}$$

where τ represents the average time.

According to (5), the slopes and intercepts of the Allan variance can be fitted to obtain the estimates of (σ_0^2, σ_1^2, σ_2^2).

2) *Frequency Domain:* In the frequency domain, the statistical characteristics of atomic clock noise can be characterized in terms of the PSD. When only WFM, RWFM, and WPM are involved, the PSD of the frequency fluctuations can be expressed as follows [35]:

$$S_y(f) = h_{-2}f^{-2} + h_0 f^0 + h_2 f^2, (0 < f < f_h) \tag{6}$$

where $S_y(f)$ represents the PSD of the frequency fluctuations in units of 1/Hz; f is the Fourier frequency; h_α is the noise coefficient; α represents the power law spectral index, where $\alpha = 2$, 0, and -2 correspond to WPM, WFM, and RWFM, respectively; and f_h is the high-frequency cutoff.

Because the instantaneous time is the integral of the instantaneous frequency, as shown in (4), the PSD of the time fluctuations can be expressed as [35]

$$\begin{aligned} S_x(f) &= \frac{1}{(2\pi f)^2} S_y(f) \\ &= \frac{1}{4\pi^2}\left(h_{-2}f^{-4} + h_0 f^{-2} + h_2 f^0\right), (0 < f < f_h) \end{aligned} \tag{7}$$

where $S_x(f)$ represents the PSD of the time fluctuations in units of s^2/Hz.

According to Riley [35], the Allan variance, which involves only WPM, WFM, and RWFM, can be expressed in terms of the noise coefficients as follows:

$$\sigma_y^2(\tau) = h_2\frac{3f_h}{(2\pi\tau)^2} + h_0\frac{1}{2\tau} + h_{-2}\frac{(2\pi)^2}{6}\tau. \tag{8}$$

Comparing (5) and (8), the relationships between the noise coefficients (h_2, h_0, h_{-2}) and diffusion coefficients (σ_0^2, σ_1^2, σ_2^2) are

$$\begin{cases} \sigma_0^2 = h_2\frac{f_h}{4\pi^2} \\ \sigma_1^2 = \frac{h_0}{2} \\ \sigma_2^2 = h_{-2}2\pi^2 \end{cases}. \tag{9}$$

To summarize, according to (6) and (7), the slopes and intercepts of the PSDs can be fitted to obtain the estimates of h_{-2}, h_0, and h_2; Then, according to (9), the estimates of (σ_0^2, σ_1^2, σ_2^2) can be obtained.

D. $\boldsymbol{Q}$ and R

The process noise vector in (3) is denoted by $\boldsymbol{J}_k$ as [2], [3], [4], [32], [33], [34]

$$\boldsymbol{J}_k = \begin{bmatrix} \sigma_1\left[W_1\left(t_{k+1}\right) - W_1\left(t_k\right)\right] + \sigma_2\int_{t_k}^{t_{k+1}}\left[W_2(s) - W_2(t_k)\right]\mathrm{d}s \\ \sigma_2\left[W_2\left(t_{k+1}\right) - W_2\left(t_k\right)\right] \end{bmatrix}. \tag{10}$$

The observation noise in (2) is denoted by w_k as [3], [4]

$$w_k = \sigma_0\xi_0\left(t_k\right). \tag{11}$$

The process noise covariance ($\boldsymbol{Q}$) of $\boldsymbol{J}_k$ appears as [2], [3], [4], [32], [33], [34]

$$\boldsymbol{Q} = \mathrm{cov}\left[\boldsymbol{J}_k, \boldsymbol{J}_k^T\right] = \begin{bmatrix} \sigma_1^2 T + \frac{1}{3}\sigma_2^2 T^3 & \frac{1}{2}\sigma_2^2 T^2 \\ \frac{1}{2}\sigma_2^2 T^2 & \sigma_2^2 T \end{bmatrix}. \tag{12}$$

The observation noise variance (R) of w_k appears as [3], [4]

$$R = \mathrm{var}\left[w_k\right] = \sigma_0^2. \tag{13}$$

In (12) and (13), cov[.] and var[.] represent the covariance and variance, respectively.

Substituting ($\sigma_0^2, \sigma_1^2, \sigma_2^2$) values into (12) and (13) yields $\boldsymbol{Q}$ and R values.

E. Basic KF Principle of Estimating Atomic Clock States

From (2), (3), (10), and (11), the process and observation equations can be rewritten in matrix form as

$$\begin{cases} \boldsymbol{s}_{k+1} = \boldsymbol{\phi}\boldsymbol{s}_k + \boldsymbol{J}_k \\ z_k = \mathbf{H}\boldsymbol{s}_k + w_k \end{cases} \tag{14}$$

where $\boldsymbol{s}_k = [x_k, y_k]^{\mathrm{T}}$, and subscript k denotes the kth time; that is, $x(t_k)$ and $y(t_k)$ in (3) and $z(t_k)$ in (2) are abbreviated as x_k, y_k, and z_k in (14), respectively; matrix $\boldsymbol{\phi} = \left[\begin{smallmatrix} 1 & T \\ 0 & 1 \end{smallmatrix}\right]$ is the transfer matrix for (3); and matrix $\mathbf{H} = [1\ 0]$ is the transfer matrix for (2).

The KF process can be described by the following five classic equations [3], [4], [18]:

$$\hat{\boldsymbol{s}}_{k,k-1} = \boldsymbol{\phi}\hat{\boldsymbol{s}}_{k-1,k-1} \tag{15}$$

$$\boldsymbol{P}_{k,k-1} = \boldsymbol{\phi}\boldsymbol{P}_{k-1,k-1}\boldsymbol{\phi}^{\mathbf{T}} + \boldsymbol{Q} \tag{16}$$

$$\boldsymbol{K}_k = \boldsymbol{P}_{k,k-1}\mathbf{H}^{\mathbf{T}}\left(\mathbf{H}\boldsymbol{P}_{k,k-1}\mathbf{H}^{\mathbf{T}} + R\right)^{-1} \tag{17}$$

$$\hat{\boldsymbol{s}}_{k,k} = \hat{\boldsymbol{s}}_{k,k-1} + \boldsymbol{K}_k\left(z_k - \mathbf{H}\hat{\boldsymbol{s}}_{k,k-1}\right) \tag{18}$$

$$\boldsymbol{P}_{k,k} = \left(\mathbf{I} - \boldsymbol{K}_k\mathbf{H}\right)\boldsymbol{P}_{k,k-1} \tag{19}$$

where z_k, $\boldsymbol{\phi}$, and $\mathbf{H}$ have the same definitions as those in (2) and (3); $\mathbf{I}$ is the 2×2 identity matrix; $\hat{\boldsymbol{s}}_{k,k}$ is the estimate of $\boldsymbol{s}_k$ at time t_k; $\hat{\boldsymbol{s}}_{k,k-1}$ represents the one-step prediction of $\boldsymbol{s}_{k-1}$ at time t_k; $\boldsymbol{K}_k$ is the 2×1 Kalman gain; and $\boldsymbol{P}_{k,k}$ and $\boldsymbol{P}_{k,k-1}$ represent the 2×2 estimate and prediction covariance, respectively, for $\hat{\boldsymbol{s}}_{k,k}$ and $\hat{\boldsymbol{s}}_{k,k-1}$.

By operating the KF according to (15)–(19), the estimates of the two-state $\hat{\boldsymbol{s}}_{k,k} = [\hat{x}_k \quad \hat{y}_k]^{\mathrm{T}}$ can be obtained. The input into the KF is z_k, and the outputs are $\hat{x}_k$ and $\hat{y}_k$.

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III. NEW METHOD

This section describes the proposed method in detail. Section III-A introduces the core idea of this method. Section III-B presents the derivation of the transfer functions for the LTI systems. Section III-C provides an analysis of the derivation process.

A. Core Idea

The core idea of the developed method can be summarized as follows.

1) The KF in the steady state can be made equivalent to three LTI systems by deriving the transfer functions of the KF.

The input of the KF is a series of discrete-time data $\{z_k\}$ (i.e., $z(t_0)$, $z(t_1)$, $z(t_2)$, $\ldots$). The outputs are three series of discrete-time data: $\{\hat{x}_k\}$ ($\hat{x}(t_0)$, $\hat{x}(t_1)$, $\hat{x}(t_2)$, $\ldots$), $\{\hat{y}_k\}$ ($\hat{y}(t_0)$, $\hat{y}(t_1)$, $\hat{y}(t_2)$, $\ldots$), and $\{\hat{y}_{int,k}\}$ ($\hat{y}_{int}(t_0)$, $\hat{y}_{int}(t_1)$, $\hat{y}_{int}(t_2)$, $\ldots$). Among them, $\hat{y}_{int,k}$ is the integral of $\hat{y}_k$, which is used to determine the theoretical performance of $\hat{y}_k$ indirectly.

As we know, the Z-transform is a useful mathematical tool for a discrete-time system. Denote the Z-transforms of z_k, $\hat{x}_k$, $\hat{y}_k$, and $\hat{y}_{int,k}$ as $Z(z)$, $X(z)$, $Y(z)$, and $Y_{INT}(z)$, respectively. The relationships among them in the Z domain appear as

$$\begin{cases} X(z) = Hx(z)Z(z) \\ Y(z) = Hy(z)Z(z) \\ Y_{INT}(z) = Hy_{int}(z)Z(z) \end{cases} \tag{20}$$

where $Hx(z)$ represents the transfer function between the KF input z_k and output $\hat{x}_k$, $Hy(z)$ denotes that between the KF input z_k and output $\hat{y}_k$, and $Hy_{int}(z)$ indicates that between z_k and $\hat{y}_{int,k}$.

According to the derivation process in Section III-B, when the KF is in the steady state, the transfer functions all represent LTI systems, which are completely determined by the KF model and chosen $\boldsymbol{Q}$ and R values, independent of actual observation characteristics.

Therefore, the KF in the steady state can be treated as equivalent to three LTI systems: $Hx(z)$, $Hy(z)$, and $Hy_{int}(z)$. The theoretical performance of the steady-state KF can be determined from the LTI system perspective.

2) The theoretical performance of each term in the KF estimates can be accurately and independently determined by using the superposition and homogeneity of the equivalent LTI systems.

Fig. 1 describes the input and outputs of the equivalent LTI systems, where the input is z_k and the outputs are $\hat{x}_k$, $\hat{y}_k$, and $\hat{y}_{int,k}$. Assume that the actual KF input z_k consists of different terms, such as $z_k = A(t_k) + B(t_k) + C(t_k) + D(t_k)\ldots$, which may be inconsistent with the KF model (14). For the outputs, the corresponding terms in $\hat{x}_k$, $\hat{y}_k$, and $\hat{y}_{int,k}$ are denoted by the subscripts x, y, and y_{int}, respectively.

Because LTI systems satisfy superposition and homogeneity [36], [37], [38], [39], each output term corresponding to its input term can be determined accurately and independently, including the terms that are not included in the KF model (14).

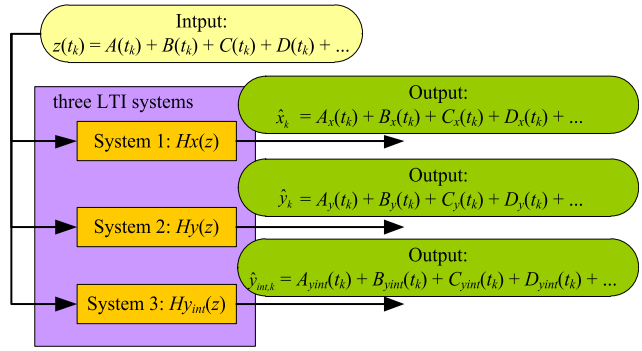

Fig. 1. Input and outputs of the LTI systems.

B. Derivation of the LTI Systems

This section presents the detailed derivation of the transfer functions of the steady-state KF and equivalent LTI systems.

Because the system in (14) is observable, the KF will reach a steady state, and $\boldsymbol{P}_{k,k}$, $\boldsymbol{P}_{k,k-1}$, and $\boldsymbol{K}_k$ all converge [36]. The steady-state matrices of $\boldsymbol{P}_{k,k}$, $\boldsymbol{P}_{k,k-1}$, and $\boldsymbol{K}_k$ are denoted as $\boldsymbol{Ps}$, $\boldsymbol{Ps}^-$, and $\boldsymbol{Ks}$, respectively. The gains in the first and second rows of the 2×1 $\boldsymbol{Ks}$ matrix are denoted as Ks_1 and Ks_2, respectively.

When the KF is in steady state, substituting (15) into (18), where $\hat{\boldsymbol{s}}_{k,k} = [\hat{x}_k \quad \hat{y}_k\ ]^{\mathrm{T}}$, gives

$$\begin{cases} \hat{x}_k = \hat{x}_{k-1} + \hat{y}_{k-1}T + Ks_1\left(z_k - \hat{x}_{k-1} - \hat{y}_{k-1}T\right) \\ \hat{y}_k = \hat{y}_{k-1} + Ks_2\left(z_k - \hat{x}_{k-1} - \hat{y}_{k-1}T\right) \end{cases}. \tag{21}$$

The innovation is denoted as

$$v_k = \left(z_k - \hat{x}_{k-1} - \hat{y}_{k-1}T\right). \tag{22}$$

Substituting (22) into (21) yields

$$\begin{cases} \hat{x}_k = \hat{x}_{k-1} + \hat{y}_{k-1}T + Ks_1 v_k \\ \hat{y}_k = \hat{y}_{k-1} + Ks_2 v_k \end{cases}. \tag{23}$$

In the Z-domain, z^{-1} means a time shift (or delay) in the time domain. Because $X(z)$ represents the Z-transform of $\hat{x}_k$, the Z-transform of $\hat{x}_{k-1}$ is $z^{-1}X(z)$. Thus, in the Z-domain, (23) can be rewritten as

$$\begin{cases} X(z) = z^{-1}X(z) + Tz^{-1}Y(z) + Ks_1V(z) \\ Y(z) = z^{-1}Y(z) + Ks_2V(z) \end{cases} \tag{24}$$

where $V(z)$ represents the Z-transform of v_k.

In the Z-domain, (22) can be rewritten as

$$V(z) = Z(z) - z^{-1}X(z) - Tz^{-1}Y(z). \tag{25}$$

From (24)

$$X(z) = \left[\frac{Ks_1}{1-z^{-1}} + \frac{Ks_2Tz^{-1}}{\left(1-z^{-1}\right)^2}\right]V(z) \tag{26}$$

$$Y(z) = \frac{Ks_2}{1-z^{-1}}V(z). \tag{27}$$

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By substituting (26) and (27) into (25)

$$\left[1-Ks_1+\frac{Ks_1}{1-z^{-1}}+\frac{Ks_2Tz^{-1}}{\left(1-z^{-1}\right)^2}\right]V(z)=Z(z). \quad (28)$$

Denote that

$$G(z)=\frac{1}{1-Ks_1}\left[\frac{Ks_1}{1-z^{-1}}+\frac{Ks_2Tz^{-1}}{\left(1-z^{-1}\right)^2}\right]. \quad (29)$$

Using (26)–(29) yields

$$X(z)=Hx(z)Z(z)=\frac{G(z)}{1+G(z)}Z(z)$$
$$=\frac{Ks_1\left(1-z^{-1}\right)+Ks_2Tz^{-1}}{(1-Ks_1)\left(1-z^{-1}\right)^2+Ks_1\left(1-z^{-1}\right)+Ks_2Tz^{-1}}Z(z) \quad (30)$$

$$Y(z)=Hy(z)Z(z)=\frac{Ks_2}{(1-Ks_1)\left(1-z^{-1}\right)}\frac{1}{1+G(z)}Z(z)$$
$$=\frac{Ks_2\left(1-z^{-1}\right)}{(1-Ks_1)\left(1-z^{-1}\right)^2+Ks_1\left(1-z^{-1}\right)+Ks_2Tz^{-1}}Z(z). \quad (31)$$

From (30), $Hx(z)$ appears as

$$Hx(z)=\frac{Ks_1\left(1-z^{-1}\right)+Ks_2Tz^{-1}}{(1-Ks_1)\left(1-z^{-1}\right)^2+Ks_1\left(1-z^{-1}\right)+Ks_2Tz^{-1}}. \quad (32)$$

From (31), $Hy(z)$ appears as

$$Hy(z)=\frac{Ks_2\left(1-z^{-1}\right)}{(1-Ks_1)\left(1-z^{-1}\right)^2+Ks_1\left(1-z^{-1}\right)+Ks_2Tz^{-1}}. \quad (33)$$

In some cases, particularly when analyzing the noise characteristics of $\hat{y}_k$, it is more intuitive to analyze the integral of $\hat{y}_k$ ($\hat{y}_{int,k}$). Note that

$$\hat{y}_{int,k}=\hat{y}_{int,k-1}+\hat{y}_{k-1}T. \quad (34)$$

In the Z-domain, (34) can be rewritten as

$$Y_{INT}(z)=z^{-1}Y_{INT}(z)+z^{-1}TY(z). \quad (35)$$

Thus, the relationship between $\hat{y}_{int,k}$ and $\hat{y}_k$ appears as

$$Y_{INT}(z)=\frac{Tz^{-1}}{1-z^{-1}}Y(z)=Int(z)Y(z) \quad (36)$$

where $Int(z)=\frac{Tz^{-1}}{1-z^{-1}}$ is the transfer function of the integrator.

Consequently, $Hy_{int}(z)$ appears as

$$Hy_{int}(z)=Hy(z)Int(z)$$
$$=\frac{Ks_2Tz^{-1}}{(1-Ks_1)\left(1-z^{-1}\right)^2+Ks_1\left(1-z^{-1}\right)+Ks_2Tz^{-1}}. \quad (37)$$

To date, the transfer functions, $Hx(z)$, $Hy(z)$, and $Hy_{int}(z)$, of the steady-state KF have been obtained.

Because coefficients Ks_1 and Ks_2 converge to constants when the KF reaches the steady state, the transfer functions all represent LTI systems. Therefore, the steady-state KF can be treated as equivalent to three LTI systems: $Hx(z)$, $Hy(z)$, and $Hy_{int}(z)$.

C. Derivation Analysis

Through the five KF equations, (15)–(19), the derivation process can be analyzed in depth to understand these LTI systems more fully.

1) Equations (15) and (18) determine the structures of the LTI systems.

These LTI systems can be obtained by substituting (15) into (18) through a series of derivations, and their structures are shown in (32), (33), and (37). Equations (15) and (18) correspond to the KF model (14); therefore, the KF model (14) completely determines the structures of the LTI systems.

2) Equations (16), (17), and (19) determine the coefficients of the LTI system.

By combining (16), (17), and (19), Ks_1 and Ks_2 values can be solved for the given $\boldsymbol{Q}$ and R values. Ks_1 and Ks_2 values correspond to the coefficients of the LTI systems. Therefore, $\boldsymbol{Q}$ and R values completely determine these coefficients.

To conclude, the equivalent LTI systems (structures and coefficients) of the classic two-state KF in the steady state are completely determined by the given KF model (14) and $\boldsymbol{Q}$ and R values and are independent of actual observations.

In the following, when we use the classic two-state KF to estimate atomic clock states, regardless of how the actual observations behave, each term in the estimates can be independently determined from the equivalent LTI system perspective when the KF reaches the steady state.

IV. ACTUAL OBSERVATION AND THREE SUBMETHODS

Section IV-A introduces the actual observation model used in this study, which involves three kinds of terms: polynomial, periodic fluctuation, and noise terms. Then, three submethods are proposed to deal with the three kinds of terms. Sections IV-B, IV-C, and IV-D describe the submethods for the polynomial, periodic fluctuation, and noise terms, respectively.

Finally, for the deterministic (polynomial and periodic fluctuation) terms, the analytical expressions for the corresponding terms in the KF estimates are derived, and for the stochastic (noise) term, the noise filtering performance is accurately determined in the frequency domain.

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A. Actual Observation

As noted in Section I, sometimes periodic fluctuations appear in observations. In these cases, the actual observations do not agree with the KF model (14). To demonstrate that the new method can effectively determine the theoretical KF performance in these cases, a periodic fluctuation term is introduced into the actual observation, which is used as an example for the following analysis and expressed as

$$\begin{aligned} z_k &= A(t_k) + B(t_k) + C(t_k) + D(t_k) \\ &= x_0 + y_0 t_k + \frac{1}{2} d t_k^2 + A_0 \sin\left(2\pi f_p t_k + \varphi_0\right) \\ &\quad + \sigma_1 W_1(t_k) + \sigma_2 \int_0^{t_k} W_2(s)\mathrm{d}s + \sigma_0 \xi_0(t_k) \end{aligned} \tag{38}$$

where $A(t_k) + B(t_k) = x_0 + y_0 t_k + (1/2) d t_k^2$ ($A(t_k) = x_0 + y_0 t_k$ and $B(t_k) = (1/2) d t_k^2$) is the polynomial term; $C(t_k) = A_0 \sin(2\pi f_p t_k + \varphi_0)$ is the introduced periodic fluctuation term, where A_0, f_p, and φ_0 indicate the amplitude, frequency, and initial phase difference, respectively; and $D(t_k)$ is the noise term that involves clock noises (WFM and RWFM) and observation noise (WPM).

Comparing the actual observation (38) with the KF observation (2), (38) superimposes a periodic fluctuation term $C(t_k)$ onto the KF observation (2) and is thus inconsistent with the KF observation (2) and KF method (14).

The example of a hydrogen maser observation is used to describe the submethods.

The observation is the real-time difference between a ground hydrogen maser and a time reference measured by a time interval counter, which is internal nonpublic data. The time reference is synchronized with coordinated universal time and more stable than the hydrogen maser. Through analysis in advance, the hydrogen maser observation is consistent with (38) and can be expressed as $z_k = A(t_k) + B(t_k) + C(t_k) + D(t_k)$. By means of the approach in [4], the corresponding values of the hydrogen maser observation can be estimated in advance as (σ_0^2, σ_1^2, σ_2^2, d, A_0, f_p) = (1.3×10^{-22} s^2, 3.7×10^{-26} s, 7.0×10^{-34} s^{-1}, -2.41×10^{-20} s^{-1}, 2.34×10^{-10} s, 1/86 400 Hz). Substituting (σ_0^2, σ_1^2, σ_2^2) values into (12) and (13) yields $\boldsymbol{Q}$ and R values. Then, by combining (16), (17), and (19), (Ks_1, Ks_2) values are obtained as (0.1294, 1.6771×10^{-5} s^{-1}). By substituting (Ks_1, Ks_2) values into (32), (33), and (37), the transfer functions of the equivalent LTI systems for this example appear as

$$Hx(z) = \frac{0.1294 z^2 - 0.1284 z}{z^2 - 1.8696 z + 0.8706} \tag{39}$$

$$Hy(z) = \frac{1.6771 \times 10^{-5} z^2 - 1.6771 \times 10^{-5} z}{z^2 - 1.8696 z + 0.8607} \tag{40}$$

$$Hy_{int}(z) = \frac{1.0063 \times 10^{-2} z}{z^2 - 1.8696 z + 0.8607}. \tag{41}$$

In the following, the LTI systems in (39)–(41) will be used as examples to discuss the submethods.

For the actual observation in (38), the frequency drift and periodic fluctuation also contribute to the Allan variance. In this case, the Allan variance is expressed as [4]

$$\begin{aligned} \sigma_y^2(\tau) &= \frac{3\sigma_0^2}{\tau^2} + \frac{\sigma_1^2}{\tau} + \sigma_2^2 \frac{\tau}{3} + \frac{1}{2} d^2 \tau^2 \\ &\quad + \left(A_0 2\pi f_p\right)^2 \frac{\sin^4\left(\pi f_p \tau\right)}{(\tau)^2}. \end{aligned} \tag{42}$$

Comparison of (42) with (5) shows that the first three terms in (42) are the same as those in (5). These are the contributions of the WPM, WFM, and RWFM. The fourth term represents the contribution of the frequency drift, and the fifth term corresponds to the periodic fluctuation. In Section VI (experimental part), the conclusion in (42) will be used.

B. Submethod for the Polynomial Term

In Fig. 1, the output terms in $\hat{x}_k$ and $\hat{y}_k$ corresponding to the input term $A(t_k) + B(t_k)$ are $A_x(t_k) + B_x(t_k)$ and $A_y(t_k) + B_y(t_k)$, respectively.

The submethod for the polynomial term $A(t_k) + B(t_k)$ is as follows. The first step is determining the error transfer functions. Second, by using the final value theorem [36], [37], [38], [39], the steady-state tracking error values are theoretically solved. Subsequently, the analytical expressions of the output terms $A_x(t_k) + B_x(t_k)$ and $A_y(t_k) + B_y(t_k)$ are derived.

1) *Error Transfer Functions:* The estimation error of $\hat{x}_k$ is defined as

$$errx_k = z_k - \hat{x}_k \tag{43}$$

where $errx_k$ represents the estimation error between z_k and $\hat{x}_k$.

By substituting (30) into (43), in the Z-domain (43) can be rewritten as

$$ERRx(z) = Z(z) - X(z) = [1 - Hx(z)] Z(z) \tag{44}$$

where $ERRx(z)$ represents the Z-transform of $errx_k$.

From (44), the error transfer function that describes the relationship between the KF input z_k and estimation error $errx_k$, denoted as $HEx(z)$, appears as

$$\begin{aligned} HEx(z) &= 1 - Hx(z) \\ &= \frac{(1 - Ks_1)\left(1 - z^{-1}\right)^2}{(1 - Ks_1)\left(1 - z^{-1}\right)^2 + Ks_1\left(1 - z^{-1}\right) + Ks_2 T z^{-1}}. \end{aligned} \tag{45}$$

In practice, the instantaneous frequency cannot be measured directly, whereas the average frequency can be measured over a period of time. In this study, the average frequency observation obtained by the difference of z_k appears as

$$\bar{y}\bar{o}_k = \frac{z\left(t_{k+1}\right) - z(t_k)}{T} \tag{46}$$

where $\bar{y}\bar{o}_k$ represents the average frequency observation.

In the Z-domain, (46) can be expressed as

$$YO(z) = \frac{1 - z^{-1}}{T z^{-1}} Z(z) \tag{47}$$

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where $YO(z)$ represents the Z-transform of $\bar{y}\bar{o}_k$.

The estimation error of $\hat{y}_k$ is defined as

$$erry_k = \bar{y}\bar{o}_k - \hat{y}_k \tag{48}$$

where $erry_k$ represents the estimation error between $\bar{y}\bar{o}_k$ and $\hat{y}_k$.

By substituting (31) and (47) into (48), (48) can be rewritten as follows in the Z-domain:

$$\begin{aligned} ERRy(z) &= YO(z) - Y(z) \\ &= \left[\frac{1-z^{-1}}{Tz^{-1}} Z(z) - Hy(z)\right] Z(z) \end{aligned} \tag{49}$$

where $ERRy(z)$ represents the Z-transform of $erry_k$.

From (33) and (49), the error transfer function that describes the relationship between the KF input z_k and estimation error $erry_k$, denoted as $HEy(z)$, appears as

$$\begin{aligned} HEy(z) &= \frac{1-z^{-1}}{Tz^{-1}} - Hy(z) \\ &= \frac{(1-Ks_1)\left(1-z^{-1}\right)^3 + Ks_1\left(1-z^{-1}\right)^2}{(1-Ks_1)\left(1-z^{-1}\right)^2 Tz^{-1} + Ks_1\left(1-z^{-1}\right)Tz^{-1} + Ks_2 T^2 z^{-2}}. \end{aligned} \tag{50}$$

In summary, (45) and (50) are the error transfer functions.

2) *Steady-State Tracking Errors:* Based on the error transfer functions in (45) and (50) and by using the final value theorem [36], [37], [38], [39], the steady-state tracking errors can be obtained as

$$\begin{cases} Ae_x(\infty) = \lim\limits_{z\to 1}\left\{\left(1-z^{-1}\right)HEx(z)Z[A(t_k)]\right\} = 0 \\ Ae_y(\infty) = \lim\limits_{z\to 1}\left\{\left(1-z^{-1}\right)HEy(z)Z[A(t_k)]\right\} = 0 \\ Be_x(\infty) = \lim\limits_{z\to 1}\left\{\left(1-z^{-1}\right)HEx(z)Z[B(t_k)]\right\} = \frac{1-Ks_1}{Ks_2}dT \\ Be_y(\infty) = \lim\limits_{z\to 1}\left\{\left(1-z^{-1}\right)HEy(z)Z[B(t_k)]\right\} = \frac{Ks_1}{Ks_2}d \end{cases} \tag{51}$$

where $Ae_x(\infty)$ represents the steady-state tracking error in $errx_k$ caused by the $A(t_k)$ term, $Ae_y(\infty)$ represents that in $erry_k$ caused by the $A(t_k)$ term, $Be_x(\infty)$ represents that in $errx_k$ caused by the $A(t_k)$ term, $Be_y(\infty)$ represents that in $erry_k$ caused by the $B(t_k)$ term, and $Z[\cdot]$ represents the Z transform of the expression in the bracket.

3) *Analytical Expressions:* Based on the error transfer functions in (45) and (50) and the steady-state tracking errors in (51), the analytical expressions can be derived as

$$\begin{cases} A_x(t_k) = A(t_k) - Ae_x(\infty) = x_0 + y_0 t_k \\ A_y(t_k) = \frac{A(t_{k+1}) - A(t_k)}{T} - Ae_y(\infty) = y_0 \\ B_x(t_k) = B(t_k) - Be_x(\infty) = \frac{1}{2}dt_k^2 - \frac{1-Ks_1}{Ks_2}dT \\ B_y(t_k) = \frac{B(t_{k+1}) - B(t_k)}{T} - Be_y(\infty) = dt_k - d\left(\frac{Ks_1}{Ks_2} - \frac{T}{2}\right) \end{cases}. \tag{52}$$

Take $B_y(t_k)$ as an example to illustrate these conclusions. Equation (51) demonstrates that $B(t_k)$ term leads to a steady-state tracking error $Be_y(\infty)$ in $erry_k$. From the definition of $erry_k$ in (48), it can be concluded that there is a constant deviation $Be_y(\infty)$ between $\bar{y}\bar{o}_k$ and $\hat{y}_k$. Using (48) and the last equation in (51), and replacing z_k with $B(t_k)$ in (46) and (48) (where $z_k = A(t_k) + B(t_k) + C(t_k) + D(t_k)$ but only $B(t_k)$ is concerned now) as well as replacing $\hat{y}_k$ with $B_y(t_k)$ in (48), yields the last equation in (52).

4) *Conclusion:* Equation (52) gives the analytical expressions of $A_x(t_k) + B_x(t_k)$ in $\hat{x}_k$ and $A_y(t_k) + B_y(t_k)$ in $\hat{y}_k$ when the KF is in the steady state.

Comparing (52) with the clock model in (1), $A_x(t_k)$ and $A_y(t_k)$ are identical to $A(t_k)$ and $\mathrm{d}A(t_k)/\mathrm{d}t$ in (1); however, deterministic deviations appear between $B_x(t_k)$ and $B_y(t_k)$ with $B(t_k)$ and $\mathrm{d}B(t_k)/\mathrm{d}t$ in (1), respectively, which are caused by steady-state tracking errors.

The main reasons are that $Hx(z)$ and $Hy(z)$ are second-order LTI systems and that $A(t_k)$ is a first-order polynomial, whereas $B(t_k)$ is a second-order polynomial. Thus, $Hx(z)$ and $Hy(z)$ can effectively track $A(t_k)$ without steady-state tracking errors [36], [37], but steady-state tracking errors appear for both $Hx(z)$ and $Hy(z)$ when tracking $B(t_k)$ [36], [37].

If these two deterministic deviations are negligible, $B_x(t_k)$ and $B_y(t_k)$ approximate $B(t_k)$ and $\mathrm{d}B(t_k)/\mathrm{d}t$, respectively. In these cases, even if the atomic clock has a frequency drift, the KF can track states $x(t)$ and $y(t)$ effectively, and the deterministic deviations have less impact on the estimates $\hat{x}_k$ and $\hat{y}_k$; it is unnecessary to use the classic three-state KF [5], [6].

C. Submethod for the Periodic Fluctuation Term

In Fig. 1, the output terms in $\hat{x}_k$, $\hat{y}_k$, and $\hat{y}_{int,k}$ are $C_x(t_k)$, $C_y(t_k)$, and $C_{yint}(t_k)$ corresponding to the input term $C(t_k)$, respectively.

The submethod for the periodic fluctuation term $C(t_k)$ is as follows. By means of the amplitude- and phase-frequency responses of the LTI systems, the analytical expressions of the output terms $C_x(t_k)$, $C_y(t_k)$, and $C_{yint}(t_k)$ are derived. The approximate relationship between $C_y(t_k)$ and $C_{yint}(t_k)$ is also given.

1) *Analytical Expressions:* Equation (38) gives the expression of the $C(t_k)$ term, which is a sine signal. As we know, when a sine signal passes an LTI system, its amplitude and phase may change [38].

By substituting the approximation $z = e^{j2\pi fT}$ when $2\pi fT << 1$, where j represents the imaginary unit into the transfer functions $Hx(z)$, $Hy(z)$, and $Hy_{int}(z)$ in the Z-domain, the transfer functions $Hx(f)$, $Hy(f)$, and $Hy_{int}(f)$ in the frequency domain are obtained.

By means of the amplitude- and phase-frequency responses of $Hx(f)$, $Hy(f)$, and $Hy_{int}(f)$, the analytical expressions for $C_x(t_k)$, $C_y(t_k)$, and $C_{yint}(t_k)$ can be derived as

$$\begin{aligned} C_x(t_k) &= A_1(f_p)\sin\left[2\pi f_p t_k + \varphi_1(f_p)\right] \\ &= \left|Hx(f_p)\right| A_0 \sin\left[2\pi f_p t_k + \varphi_0 + \angle Hx(f_p)\right] \end{aligned} \tag{53}$$

$$\begin{aligned} C_y(t_k) &= A_2(f_p)\sin\left[2\pi f_p t_k + \varphi_2(f_p)\right] \\ &= \left|Hy(f_p)\right| A_0 \sin\left[2\pi f_p t_k + \varphi_0 + \angle Hy(f_p)\right] \end{aligned} \tag{54}$$

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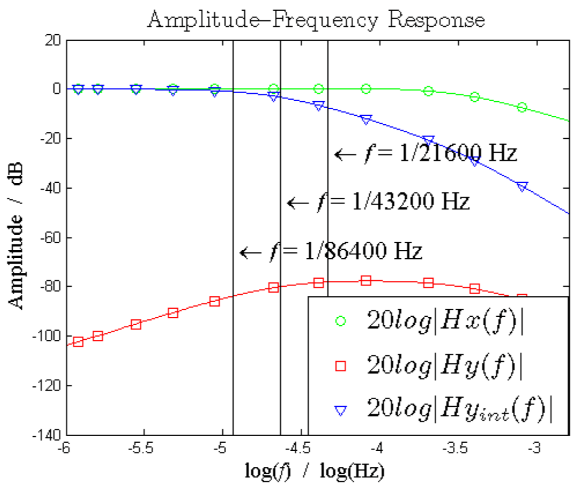

Fig. 2. Amplitude–frequency response curves.

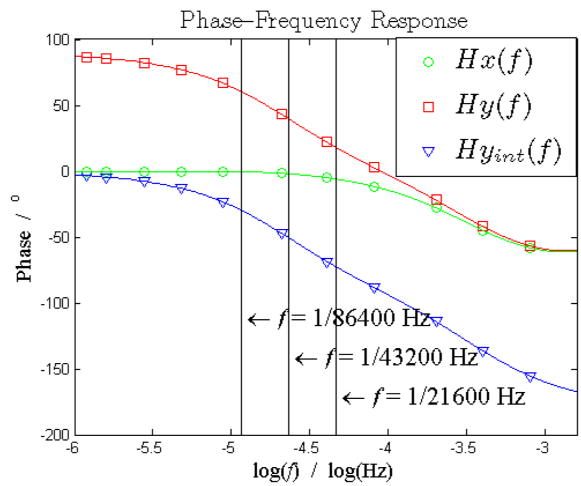

Fig. 3. Phase–frequency response curves.

$$\begin{aligned} C_{yint}(t_k) &= A_3(f_p)\sin\left[2\pi f_p t_k + \varphi_3(f_p)\right] \\ &= \left|Hy_{int}(f_p)\right| A_0 \sin\left[2\pi f_p t_k + \varphi_0 + \angle Hy_{int}(f_p)\right] \end{aligned} \tag{55}$$

where $A_1(f_p) = |Hx(f_p)|A_0$, $A_2(f_p) = |Hy(f_p)|A_0$, and $A_3(f_p) = |Hy_{int}(f_p)|A_0$, respectively, represent the amplitudes of $C_x(t_k)$, $C_y(t_k)$, and $C_{yint}(t_k)$; $\varphi_1(f_p) = \varphi_0 + \angle Hx(f_p)$, $\varphi_2(f_p) = \varphi_0 + \angle Hy(f_p)$,, and $\varphi_3(f_p) = \varphi_0 + \angle Hy_{int}(f_p)$, respectively, represent the phases of $C_x(t_k)$, $C_y(t_k)$, and $C_{yint}(t_k)$; and $\angle Hx(f_p)$, $\angle Hy(f_p)$, and $\angle Hy_{int}(f_p)$, respectively, represent the angles of $Hx(f)$, $Hy(f)$, and $Hy_{int}(f)$ at frequency f_p.

Moreover, the approximate expression for $C_y(t_k)$ can also be derived as

$$\begin{aligned} C_y(t_k) &= A_2(f_p)\sin\left[2\pi f_p t_k + \varphi_2(f_p)\right] \\ &\approx A_3(f_p)\, 2\pi f_p \sin\left[2\pi f_p t_k + \varphi_3(f_p) + \frac{\pi}{2}\right]. \end{aligned} \tag{56}$$

The detailed derivation process for (56) is provided in the appendix.

2) *Example:* The hydrogen maser observation described in Section IV-A is used as an example to show the amplitude– and phase–frequency responses of the equivalent LTI systems, whose transfer functions are expressed as (39)–(41) in the Z-domain in this example.

Fig. 2 depicts the logarithmic amplitude–frequency response curves of $Hx(f)$, $Hy(f)$, and $Hy_{int}(f)$. Fig. 3 plots the logarithmic phase–frequency response curves of $Hx(f)$, $Hy(f)$, and $Hy_{int}(f)$, where the phase is in units of radians and $180° = \pi$ rad.

Based on these responses, the amplitudes $A_1(f_p)$, $A_2(f_p)$, and $A_3(f_p)$ and phases $\varphi_1(f_p)$, $\varphi_2(f_p)$, and $\varphi_3(f_p)$ for different frequencies f_p can be theoretically determined.

For example, Table I lists $|Hx(f)|$, $|Hy(f)|$, $|Hy_{int}(f)|$, $\angle Hx(f_p)$, $\angle Hy(f_p)$, and $\angle Hy_{int}(f_p)$ values for f_p = 1/86 400, 1/43 200, and 1/21 600 Hz.

The results in Table I validate the approximation in (56), as the approximation errors are very small when substituting the values in Table I into (56).

TABLE I
Different Amplitudes and Phases

f_p (Hz)	1/86 400	1/43 200	1/21 600
$\|Hx(f)\|$	1.0127	1.0294	1.0395
$\|Hy(f)\|$ (1/s)	6.4282×10^{-5}	9.9818×10^{-5}	1.2345×10^{-4}
$\|Hy_{int}(f)\|$	0.8839	0.6863	0.4244
$\angle Hx(f_p)$ (rad)	–0.0071	–0.0338	–0.1027
$\angle Hy(f_p)$ (rad)	1.0539	0.6959	0.3179
$\angle Hy_{int}(f_p)$ (rad)	–0.5190	–0.8793	–1.2617

From Figs. 2 and 3 and Table I, the amplitudes and phases of $C_x(t_k)$ and $C_y(t_k)$ change with respect to $C(t_k)$ and $dC(t_k)/dt$, respectively, and can be accurately determined. These quantities are different for different f_p values. For example, when f_p = 1/86 400 Hz, $A_1(f_p) \approx A_0$ and $\varphi_1(f_p) \approx \varphi_0$; and then from (38) and (53), it can be concluded that $C_x(t_k) \approx C(t_k)$.

3) *Conclusion:* The analytical expressions for $C_x(t_k)$ and $C_y(t_k)$ are given by (53) and (54), respectively. The approximate expression of $C_y(t_k)$ is given by (56), which is determined indirectly through $C_{yint}(t_k)$.

Although the KF does not model $C(t_k)$, the analytical expressions for the corresponding terms, $C_x(t_k)$ and $C_y(t_k)$, in $\hat{x}_k$ and $\hat{y}_k$ can be derived theoretically. The amplitudes and phases of $C_x(t_k)$ and $C_y(t_k)$ change compared with those of $C(t_k)$ and $dC(t_k)/dt$, respectively. If $C_x(t_k) \approx C(t_k)$ or $C_y(t_k) \approx dC(t_k)/dt$, then the $C(t_k)$ term in the KF input will not cause obvious deterioration of the estimation error for $\hat{x}_k$ or $\hat{y}_k$, respectively.

D. Submethod for the Noise Term

In Fig. 1, the output terms in $\hat{x}_k$, $\hat{y}_k$, and $\hat{y}_{int,k}$ are $D_x(t_k)$, $D_y(t_k)$, and $D_{yint}(t_k)$ corresponding to the input term $D(t_k)$, respectively.

The submethod for the noise term $D(t_k)$ is as follows. The equivalent LTI systems are treated as filters in the frequency domain. By means of the amplitude–frequency responses, the theoretical expressions for determining the PSDs of the output terms $D_x(t_k)$, $D_y(t_k)$, and $D_{yint}(t_k)$ are

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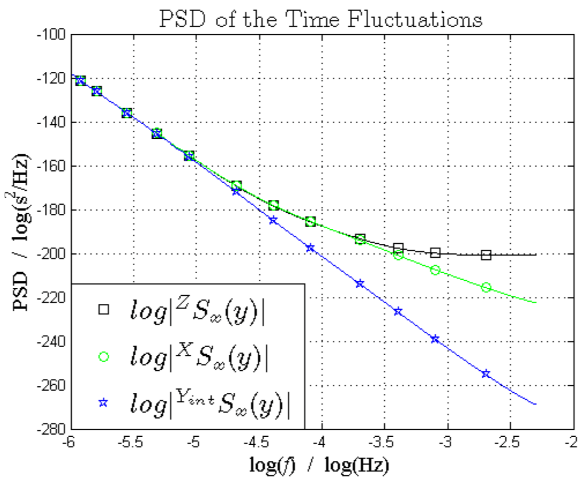

Fig. 4. ${}^{Z}S_x(f)$, ${}^{X}S_x(f)$, and ${}^{Y_{int}}S_x(f)$ curves.

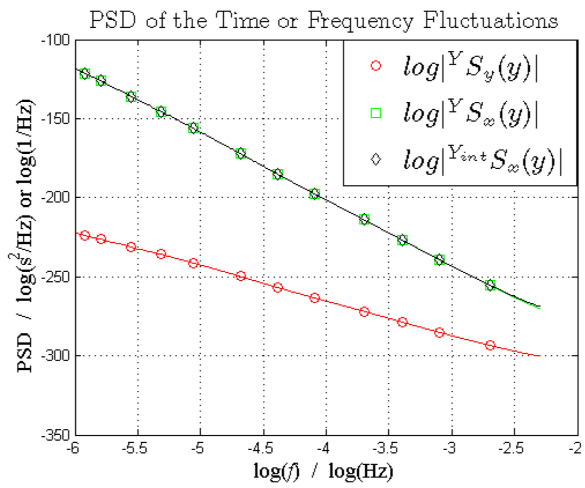

Fig. 5. ${}^{Y}S_y(f)$, ${}^{Y}S_x(f)$, and ${}^{Y_{int}}S_x(f)$ curves.

given. Consequently, the noise filtering performance of the steady-state KF is accurately determined in the frequency domain. An example is used to illustrate the submethod.

1) *Theoretical Expressions for Solving the PSDs:* As noted in Section II, $S_y(f)$ represents the PSD of the frequency fluctuations, and $S_x(f)$ represents the PSD of the time fluctuations. It is known that $\{z_k\}$, $\{\hat{x}_k\}$, and $\{\hat{y}_{int,k}\}$ are all time series, whereas $\{\hat{y}_k\}$ is a frequency series. In this study, the PSDs of the time fluctuations of $D(t_k)$, $D_x(t_k)$, and $D_{yint}(t_k)$ are denoted as ${}^{Z}S_x(f)$, ${}^{X}S_x(f)$, and ${}^{Y_{int}}S_x(f)$, and the PSD of the frequency fluctuations of $D_y(t_k)$ is denoted as ${}^{Y}S_y(f)$.

The theoretical expressions for solving the PSDs of $D_x(t_k)$, $D_y(t_k)$, and $C_{yint}(t_k)$ appear as [36], [37], [38], [39]

$$ {}^{X}S_x(f) = \left|Hx\left(f\right)\right|^2 {}^{Z}S_x(f) \tag{57}$$

$$ {}^{Y_{int}}S_x(f) = \left|Hy_{int}\left(f\right)\right|^2 {}^{Z}S_x(f) \tag{58}$$

$$ {}^{Y}S_y(f) = \left|Hy\left(f\right)\right|^2 {}^{Z}S_x(f). \tag{59}$$

The difference between (57), (58), and (59) is that the output of $Hy(f)$ is a frequency series whose PSD should be written as $S_y(f)$ rather than $S_x(f)$.

2) *Example:* The hydrogen maser observation described in Section IV-A is used as an example to show the noise filtering performances of the equivalent LTI systems, whose transfer functions are expressed as (39)–(41) in the Z-domain in this example.

Substituting $(\sigma_0^2, \sigma_1^2, \sigma_2^2)$ values into (9) and setting $f_h = 1/T$ and $T = 60$ s yields $(h_2, h_0, h_{-2}) = (3.08 \times 10^{-19}, 7.40 \times 10^{-26}, 3.55 \times 10^{-35})$. Substituting these (h_2, h_0, h_{-2}) values into (7) yields ${}^{Z}S_x(f)$. By using (57) and (58), ${}^{X}S_x(f)$ and ${}^{Y_{int}}S_x(f)$ can be obtained.

Fig. 4 plots the logarithmic curves of ${}^{Z}S_x(f)$, ${}^{X}S_x(f)$, and ${}^{Y_{int}}S_x(f)$ in this case. Based on their slopes and intercepts, ${}^{Z}S_x(f)$ incorporates WPM, WFM, and RWFM; ${}^{X}S_x(f)$ includes WFM and RWFM with the same noise levels (σ_1^2, σ_2^2); and ${}^{Y_{int}}S_x(f)$ only involves RWFM with the same noise level σ_2^2.

According to the WFM and RWFM expressions in (1), $D_x(t_k)$ can be written in the form $\sigma_1 \hat{W}_1(t_k) + \sigma_2 \int_0^{t_k} \hat{W}_2(s)\mathrm{d}s$, and $D_{yint}(t_k)$ can be expressed as $\sigma_2 \int_0^{t_k} \hat{W}_2(s)\mathrm{d}s$, where the hat symbol represents the estimation.

By using (59), ${}^{Y}S_y(f)$ can be obtained. Then, applying (7) yields ${}^{Y}S_x(f)$, which represents the PSD of the time fluctuations of the corresponding time series of the integral of $\{\hat{y}_k\}$.

Fig. 5 plots the logarithmic curves of ${}^{Y}S_y(f)$, ${}^{Y}S_x(f)$, and ${}^{Y_{int}}S_x(f)$, where the ${}^{Y_{int}}S_x(f)$ curve is identical to that in Fig. 4. In Fig. 5, the ${}^{Y}S_x(f)$ and ${}^{Y_{int}}S_x(f)$ curves coincide, indicating that their PSDs are identical and they involve the same types and levels of noise.

Note that RWFM in a frequency series $\{\hat{y}_k\}$ can be represented by a Wiener process, as shown in (1) and (2). Therefore, $D_y(t_k)$ can be expressed in the form $\sigma_2 \hat{W}_2(t_k)$.

3) *Conclusion:* This section provides theoretical expressions for solving the PSDs of $D_x(t_k)$, $D_y(t_k)$, and $D_{yint}(t_k)$, as shown in (57)–(59). An example is used to describe the theoretical performances of $D_x(t_k)$ and $D_y(t_k)$ from the filter perspective in the frequency domain. In this example, the $(\sigma_0^2, \sigma_1^2, \sigma_2^2)$ values are known in advance; thus, $\boldsymbol{Q}$ and R values are considered completely consistent with the observation characteristics. In this case, the following conclusions can be drawn.

First, when $D(t_k)$ involves WPM, WFM, and RWFM with noise levels $(\sigma_0^2, \sigma_1^2, \sigma_2^2)$, $D_x(t_k)$ only includes WFM and RWFM with the same noise levels (σ_1^2, σ_2^2), and $D_y(t_k)$ and $D_{yint}(t_k)$ only involve RWFM with the same noise level σ_2^2.

Second, from the filter perspective in the frequency domain, $Hx(f)$ filters out WPM and retains WFM and RWFM as a low-pass filter, and $Hy_{int}(f)$ filters out WPM and WFM and retains RWFM as a low-pass filter. It is more intuitive to determine the theoretical performance of $D_y(t_k)$ indirectly through $D_{yint}(t_k)$, with the same conclusions.

To summarize, the KF performs as a filter for both $\hat{x}_k$ and $\hat{y}_k$. For $\hat{x}_k$, the KF filters out the WPM in z_k; thus, $\hat{x}_k$ only involves the WFM and RWFM. For $\hat{y}_k$, the KF filters out the WPM and WFM in z_k; thus, $\hat{y}_k$ only involves the RWFM. Consequently, $D_x(t_k)$ can be expressed

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as $\sigma_1\hat{W}_1(t_k)+\sigma_2\int_0^{t_k}\hat{W}_2(s)\mathrm{d}s$, $D_y(t_k)$ can be written as $\sigma_2\hat{W}_2(t_k)$, and $D_{yint}(t_k)$ can be expressed as $\sigma_2\int_0^{t_k}\hat{W}_2(s)\mathrm{d}s$. The $D_x(t_k)$ and $D_y(t_k)$ terms in $\hat{x}_k$ and$\hat{y}_k$, respectively, are completely consistent with the corresponding terms in the atomic clock model in (1).

V. COMPLETE AND EXTENDED CONCLUSION

A. Complete Conclusions

Combining the conclusions in Section IV, when the KF is in the steady state and the $\boldsymbol{Q}$ and R values are consistent with the actual noise characteristics, for the KF input [observation z_k, as shown in (38)], the KF outputs (estimates $\hat{x}_k$ and $\hat{y}_k$) can be derived as

$$\begin{cases}\hat{x}_k=A_x(t_k)+B_x(t_k)+C_x(t_k)+D_x(t_k)\\ =x_0+y_0t_k+\frac{1}{2}dt_k^2-\frac{1-Ks_1}{Ks_2}dT\\ +A_1(f_p)\sin\left[2\pi f_pt_k+\varphi_1(f_p)\right]+\sigma_1\hat{W}_1(t_k)\\ +\sigma_2\int_0^{t_k}\hat{W}_2(s)\mathrm{d}s\\ \hat{y}_k=A_y(t_k)+B_y(t_k)+C_y(t_k)+D_y(t_k)\\ =y_0+dt_k-d\left(\frac{Ks_1}{Ks_2}-\frac{T}{2}\right)\\ +A_2(f_p)\sin\left[2\pi f_pt_k+\varphi_2(f_p)\right]+\sigma_2\hat{W}_2(t_k)\end{cases}. \tag{60}$$

In (60), the deterministic deviations caused by the steady-state tracking errors of the frequency drift and the amplitude and phase changes in the output periodic fluctuation terms are theoretically determined and should attract universal attention in the future.

Although (60) determines the noise characteristics of $\hat{x}_k$ and $\hat{y}_k$ for which $D_x(t_k)$ involves WFM and RWFM with the same noise levels (σ_1^2,σ_2^2) and $D_y(t_k)$ involves only RWFM with the same noise level σ_2^2, the paths of the stochastic processes $\hat{W}_1(t_k)$ and $\hat{W}_2(t_k)$ are not arbitrary and need to be constrained by the conclusions of traditional methods.

Assume that z_k only involves the stochastic term, which is equivalent to the special case that $A(t_k)=B(t_k)=C(t_k)=0$ and, thus, $z_k=D(t_k)$. In this case, z_k is consistent with the KF model (14). When the $\boldsymbol{Q}$ and R values are consistent with actual noise characteristics, the conclusions of traditional methods are obtained that the mean square estimation errors of both $\hat{x}_k$ and $\hat{y}_k$ reach the minimum [3], [18], [22], [23]

$$\begin{cases}E\left[(x_k-\hat{x}_k)^2\right]\\ =E\left[\left(\sigma_1W_1(t_k)+\sigma_2\int_0^{t_k}W_2(s)\mathrm{d}s-\sigma_1\hat{W}_1(t_k)\right.\right.\\ \left.\left.\quad-\sigma_2\int_0^{t_k}\hat{W}_2(s)\mathrm{d}s\right)^2\right]\\ =Ps(1,1)\rightarrow\min\\ E\left[\left(y_k-\hat{y}_k\right)^2\right]=E\left[\left(\sigma_2W_2(t_k)-\sigma_2\hat{W}_2(t_k)\right)^2\right]\\ =Ps(2,2)\rightarrow\min\end{cases} \tag{61}$$

where $\boldsymbol{Ps}$ is the steady-state matrix of $\boldsymbol{P}_{k,k}$ in (15)–(19) and $Ps(i,j)$ are the values in the ith row and jth column of $\boldsymbol{Ps}$. As described in Section III-B, $\boldsymbol{Ps}$ is determined by $\boldsymbol{Q}$ and R and can be obtained by solving (16), (17), and (19).

Because LTI systems satisfy superposition and homogeneity properties, the conclusions in (61) are also valid for the case $z_k=A(t_k)+B(t_k)+C(t_k)+D(t_k)$.

To summarize, (60) and (61) are the complete conclusions that combine the new and traditional methods for the theoretical performances of $\hat{x}_k$ and $\hat{y}_k$, when $z_k=A(t_k)+B(t_k)+C(t_k)+D(t_k)$ and the $\boldsymbol{Q}$ and R values agree with the actual noise characteristics.

B. Conclusion Extensions

Other conclusions can be extended based on (60) for this example, $z_k=A(t_k)+B(t_k)+C(t_k)+D(t_k)$, such as the following.

1) Determining the theoretical performances of the estimates when the $\boldsymbol{Q}$ and R values are not consistent with the actual noise characteristics.

Changing the $\boldsymbol{Q}$ and R values is equivalent to changing the LTI system coefficients Ks_1 and Ks_2.

For deterministic terms $A(t_k)$, $B(t_k)$, and $C(t_k)$, the analytical expressions of their corresponding output terms in (60) can also be derived in cases of different Ks_1 and Ks_2 values when the $\boldsymbol{Q}$ and R values are inconsistent with the actual noise characteristics.

For the stochastic term $D(t_k)$, the PSDs of the corresponding output terms, $D_x(t_k)$ and $D_y(t_k)$, can still be determined theoretically in the future based on the proposed submethod of treating the KF as a filter in the frequency domain.

However, the KF may remove more or less noise in this case. Thus, the noise characteristics of $D_x(t_k)$ and $D_y(t_k)$ may no longer be consistent with the corresponding expressions in (60) or the conclusions in Section IV-D. In addition, the mean-squared estimation errors related to the stochastic term are still obtained as the corresponding $Ps(1,1)$ and $Ps(2,2)$ for the given $\boldsymbol{Q}$ and R values, but they no longer reach the minimum. Hence, the mean-squared estimation errors related to the stochastic term are not optimal.

2) Determining the mean-squared estimation errors of the estimates when the KF model does not agree with actual observation characteristics.

The deterministic and stochastic terms in z_k, $\hat{x}_k$, and $\hat{y}_k$ are denoted as ${}^Dz_k$, ${}^D\hat{x}_k$, and ${}^D\hat{y}_k$ and ${}^Sz_k$, ${}^S\hat{x}_k$, and ${}^S\hat{y}_k$, respectively.

Because the deterministic and stochastic terms are independent of each other, the mean-squared estimation errors of $\hat{x}_k$ and $\hat{y}_k$ are

$$\begin{cases}E\left[(x_k-\hat{x}_k)^2\right]=E\left[\left({}^Dx_k-{}^D\hat{x}_k\right)^2\right]+E\left[\left({}^Sx_k-{}^S\hat{x}_k\right)^2\right]\\ =E\left[\left({}^Dx_k-{}^D\hat{x}_k\right)^2\right]+Ps(1,1)\\ E\left[\left(y_k-\hat{y}_k\right)^2\right]=E\left[\left({}^Dy_k-{}^D\hat{y}_k\right)^2\right]+E\left[\left({}^Sy_k-{}^S\hat{y}_k\right)^2\right]\\ =E\left[\left({}^Dy_k-{}^D\hat{y}_k\right)^2\right]+Ps(2,2)\end{cases}. \tag{62}$$

From (60), the deviations in ${}^Dx_k-{}^D\hat{x}_k$ and ${}^Dy_k-{}^D\hat{y}_k$ result from the steady-state tracking errors of the frequency drift term and the amplitude and phase changes of the periodic fluctuation terms.

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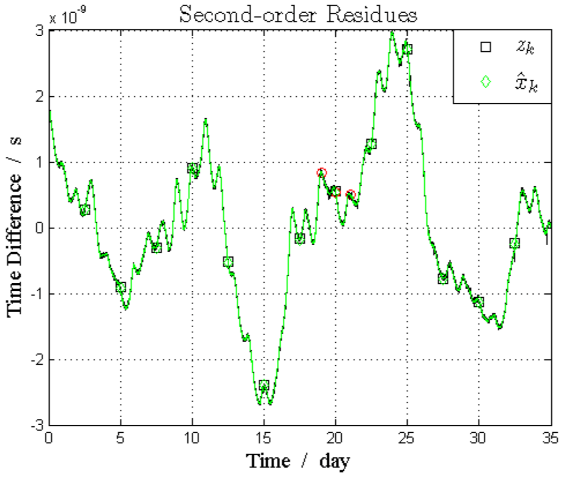

Fig. 6. Residues of z_k and $\hat{x}_k$.

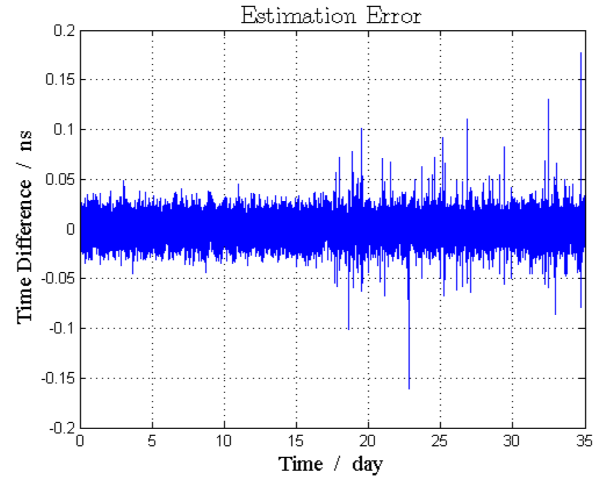

Fig. 7. Estimation error $errx_k$.

When the deviations, ${}^{D}x_k - {}^{D}\hat{x}_k$ and ${}^{D}y_k - {}^{D}\hat{y}_k$, are negligible, the mean-squared estimation errors of $\hat{x}_k$ and $\hat{y}_k$ are also approximately minimized to $Ps(1, 1)$ and $Ps(2, 2)$ according to (62). Otherwise, the mean-squared estimation errors related to each deterministic term can be solved accurately according to z_k, $\hat{x}_k$, and $\hat{y}_k$ expressions shown in (38) and (60).

In summary, this research provides a positive answer to the question raised in Section I. That is, we can theoretically determine the KF estimation and filtering performances when the KF model and its $\boldsymbol{Q}$ and R values do not completely agree with realistic observations.

VI. ACTUAL MEASURED DATA EXPERIMENT

This section uses empirical data from the same hydrogen maser described in Section IV-A to validate the theoretical conclusions. Section V-A describes the experimental data, which conform to (38). Section V-B discusses the experimental procedures and results. Section V-C analyzes the experimental results, and Section V-D summarizes the experimental conclusions.

A. Procedures and Results

The observation z_k used in this experiment is the real-time difference between the hydrogen maser and time reference, which is described in Section IV-A. The data length is 35 days, and the time interval between the data points is 60 s.

The classic two-state KF is used to obtain the estimates $\hat{x}_k$ and $\hat{y}_k$ from the observation z_k. The observation and process equations are expressed in (14). The KF is initialized according to the approach in [4], including by choosing suitable values of $\boldsymbol{Q}$ and R, initial values of $\hat{x}_k$ and $\hat{y}_k$, and $\boldsymbol{P}_{k,k}$, $\boldsymbol{P}_{k,k-1}$, and $\boldsymbol{K}_k$ in (15)–(19).

After KF initialization, the KF is run according to (15)–(19), and thus estimates $\hat{x}_k$ and $\hat{y}_k$ are obtained.

Fig. 6 plots the residuals of z_k (black) and $\hat{x}_k$ (green). Fig. 7 presents the curve for the estimation error, $errx_k = z_k - \hat{x}_k$, as defined in (43). Fig. 8 depicts the $\hat{y}_k$ curve. The red circles in Figs. 6 and 8 represent the peaks of the periodic fluctuation at the 19th, 20th, and 21st days.

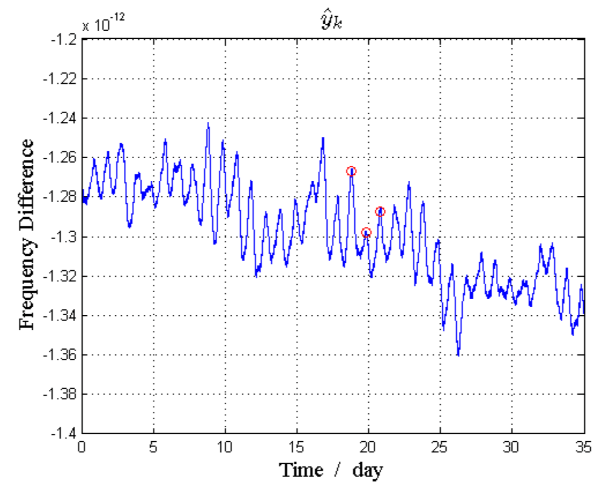

Fig. 8. Estimate $\hat{y}_k$.

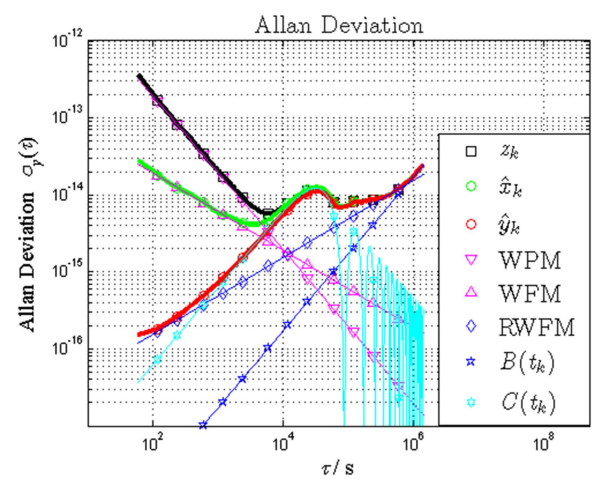

Fig. 9. Allan deviation curves.

Fig. 9 plots the Allan deviation curves for z_k (thick black), $\hat{x}_k$ (thick green), and $\hat{y}_k$ (thick red), including the contributions of the WPM (magenta with low triangles, slope = –1), WFM (magenta with up triangles, slope =

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–0.5), RWFM (blue with diamonds, slope = 0.5), frequency drift (blue with five-pointed stars, slope = 1), and periodic fluctuation (cyan) components calculated from (42). The WPM, WFM, and RWFM components are calculated according to $(\sigma_0^2, \sigma_1^2, \sigma_2^2) = (1.3 \times 10^{-22}\ \mathrm{s}^2, 3.7 \times 10^{-26}\ \mathrm{s}, 7.0 \times 10^{-34}\ \mathrm{s}^{-1})$, as shown in Section V-A, respectively. The frequency drift component is calculated according to $d = -2.41 \times 10^{-20}\ \mathrm{s}^{-1}$. The periodic fluctuation component is calculated according to $A_0/(2\pi f_p) = 1.70 \times 10^{-14}$ and $A_0 = 2.34 \times 10^{-10}$ s.

B. Results Analysis

1) *Deterministic Terms in $\hat{x}_k$:* The experimental results in Figs. 6, 7, and 9 are as follows.

First, Figs. 6 and 7 indicate that there is no obvious deterministic deviation between z_k and $\hat{x}_k$. That is, ${}^D\hat{x}_k = A_x(t_k) + B_x(t_k) + C_x(t_k) \approx {}^Dz_k = A(t_k) + B(t_k) + C(t_k)$.

According to Fig. 7, the maximum absolute value of $errx_k$ is less than 0.2 ns, and the noise characteristic of $errx_k$ approximates WPM. Statistical analysis of the data in Fig. 7 shows that the mathematical expectation is 6.44×10^{-5} ns and that the standard deviation is 0.01 ns.

Second, the Allan deviation results in Fig. 9 also suggest that the amplitudes of $C(t_k)$ and $C_x(t_k)$ approach each other.

The experimental results conform to the theoretical conclusions as follows.

First, for the polynomial term, by substituting the corresponding values of the hydrogen maser into (52), the theoretical conclusions are $A_x(t_k) = A(t_k)$ and $B_x(t_k) = B(t_k) - 7.51 \times 10^{-5}$ ns. The deterministic deviation between $B_x(t_k)$ and $B(t_k)$ is on the order of 10^{-5} ns, whereas the standard deviation of $errx_k = z_k - \hat{x}_k$ is on the order of 10^{-2} ns. Thus, the deterministic deviation can be neglected.

Second, for the periodic fluctuation term, by substituting the corresponding values of the hydrogen maser into (52), the theoretical conclusion is

$$C_x(t_k) = 1.01A_0 \sin\left[2\pi f_p t_k + \varphi_0 - 0.0071\right]. \tag{63}$$

The phase in (63) is in units of radians.

Comparison of $C(t_k)$ (38) with $C_x(t_k)$ (63) demonstrates that $C_x(t_k) \approx C(t_k)$. Consequently, in this experiment, the KF effectively tracks $C(t_k)$ with a small error between $C_x(t_k)$ and $C(t_k)$, although the KF does not model the $C(t_k)$ term.

2) *Deterministic Terms in $\hat{y}_k$:* The experimental results in Figs. 8 and 9 are as follows.

First, Fig. 8 indicates that the deterministic terms of $\hat{y}_k$ involve polynomial and periodic fluctuation terms.

The polynomial fitting result of z_k is $A(t_k) + B(t_k) \approx -0.20 - 1.26 \times 10^{-12}t - 1/2 \times 2.41 \times 10^{-20}t^2$. That of $\hat{y}_k$ is $A_y(t_k) + B_y(t_k) \approx -1.26 \times 10^{-12} - 2.28 \times 10^{-20}t$. Then, the conclusion is obtained that $A_y(t_k) + B_y(t_k) \approx \mathrm{d}A(t_k)/\mathrm{d}t + \mathrm{d}B(t_k)/\mathrm{d}t$.

Second, the Allan deviation results in Fig. 9 suggest that the amplitude of $C_y(t_k)2\pi f_p$ is slightly less than that of $C(t_k)$.

The experimental results conform to the theoretical conclusions as follows.

First, for the polynomial term, by substituting the corresponding values of the hydrogen maser into (52), the theoretical conclusions are $A_y(t_k) = \mathrm{d}A(t_k)/\mathrm{d}t$ and $B_y(t_k) = \mathrm{d}B(t_k)/\mathrm{d}t - 1.85 \times 10^{-16}$. Because the deterministic deviation between $B_y(t_k)$ and $\mathrm{d}B(t_k)/\mathrm{d}t$ is on the order of 10^{-16}, it can be neglected compared with the $\hat{y}_k$ value, which is on the order of 10^{-12}, as shown in Fig. 8.

Second, for the periodic fluctuation term, by substituting the corresponding values of the hydrogen maser into (56), the theoretical conclusion is

$$C_y(t_k) \approx 0.88A_0 2\pi f_p \sin\left[2\pi f_p t_k + \varphi_0 + \frac{\pi}{2} - 0.5190\right]. \tag{64}$$

The phase in (64) is also in units of radians.

Comparing $C(t_k)$ (38) with $C_x(t_k)$ (64), the phase of $C_y(t_k)$ approximates that of $\mathrm{d}C(t_k)/\mathrm{d}t$, whereas the amplitude of $C_y(t_k)$ is 12% lower than that of $\mathrm{d}C(t_k)/\mathrm{d}t$.

3) *Stochastic Terms in $\hat{x}_k$ and $\hat{y}_k$:* Fig. 9 indicates that z_k involves WPM, WFM, and RWFM; $\hat{x}_k$ involves WFM and RWFM with the same noise levels (σ_1^2, σ_2^2); and $\hat{y}_k$ only involves RWFM with the same noise level σ_2^2. The $D_x(t_k)$ and $D_y(t_k)$ terms are consistent with the corresponding terms in (60) as $\sigma_1\hat{W}_1(t) + \sigma_2\int_0^t \hat{W}_2(s)\mathrm{d}s$ and $\sigma_2\hat{W}_2(t)$, respectively.

Figs. 6 and 8 also support the conclusions. In Fig. 8, 35 periods in 35 days can be clearly seen, where the red circles represent the peaks. Because only RWFM is involved in $C_y(t_k)$, which is far less noisy than $C(t_k)$, periodic fluctuation $C_y(t_k)$ can be clearly seen in $\hat{y}_k$ in Fig. 8. However, periodic fluctuation $C(t_k)$ is not obvious in the residual of z_k in Fig. 6 because the noise $D(t_k)$ is relatively large compared with $C(t_k)$ in the residual of z_k.

In summary, the Allan deviation results in the time domain shown in Fig. 9 are in perfect agreement with the PSD results in the frequency domain shown in Figs. 3 and 4.

C. Experimental Conclusions

In this experiment, the KF is used to obtain the estimates $\hat{x}_k$ and $\hat{y}_k$ from the observation z_k. The KF is run according to the KF model (14), and the $\boldsymbol{Q}$ and R values are obtained in advance and consistent with the observation characteristics. As discussed in Section III, in this case, the KF in the steady state is equivalent to three LTI systems (39)–(41).

The actual observation conforms to (38) as $z_k = A(t_k) + B(t_k) + C(t_k) + D(t_k)$ in this experiment. Although the actual observation does not agree with the KF model in (14), where the observation equation in (14) is expressed as $z_k = A(t_k) + B(t_k) + D(t_k)$, each term in the estimates $\hat{x}_k$ and $\hat{y}_k$ can still be theoretically determined from the LTI system perspective.

The results of $\hat{x}_k = A_x(t_k) + B_x(t_k) + C_x(t_k) + D_x(t_k)$ and $\hat{y}_k = A_y(t_k) + B_y(t_k) + C_y(t_k) + D_y(t_k)$ from this experiment support the theoretical conclusions as follows.

First, the analytical expressions of the deterministic terms ${}^D\hat{x}_k = A_x(t_k) + B_x(t_k) + C_x(t_k)$ and ${}^D\hat{y}_k = A_y(t_k) + B_y(t_k) + C_y(t_k)$ can be theoretically derived. The deterministic deviations ${}^Dz_k - {}^D\hat{x}_k$ and $d^Dz_k/dt - {}^D\hat{y}_k$ result from the steady tracking errors caused by the frequency drift and

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the amplitude and phase changes of the periodic fluctuation. In this experiment, ${}^{D}\hat{x}_k \approx {}^{D}z_k$, which means that the KF can effectively track the $\hat{x}_k$ state with less estimation error, as shown in Figs. 6 and 7. Therefore, although the KF does not model the periodic fluctuation term $C(t_k)$, the corresponding terms $C_x(t_k)$ and $C_y(t_k)$ in the estimates can be theoretically determined by the proposed method from the LTI system perspective.

Second, when the $\boldsymbol{Q}$ and R values are consistent with the actual noise characteristics, the noise terms $D_x(t_k)$ and $D_y(t_k)$ in the estimates can be expressed as $\sigma_1\hat{W}_1(t) + \sigma_2\int_0^t \hat{W}_2(s)\mathrm{d}s$ and $\sigma_2\hat{W}_2(t)$, which are consistent with the corresponding terms in the atomic clock model (1). The results also confirm that state $y(t)$ in the atomic clock model (1) represents only the random walk part of the instantaneous frequency and only involves the RWFM. That is, state $y(t)$ is different from the instantaneous frequency (4), which should attract attention.

VII. DISCUSSION

The proposed method can be applied to GNSS onboard clock estimation effectively.

As noted in Section I, the observations of the differences between GNSS onboard clocks and ground time references, which involve significant periodic fluctuations, can be obtained in real time. The KF approach and other methods, such as sequential least squares adjustment, have been used to deal with the observations [3], [4], [5], [6]. According to the research, the new method and submethods can be applied to GNSS onboard clock estimations to obtain the theoretical KF estimation performance effectively.

Moreover, the core idea of the proposed method can potentially be applied to several current and future problems, as follows.

1) Determining the theoretical performances of other KFs, such as the three-state KF [5], [6] and another two-state KF [8], [17], including the frequency and frequency drift states.

These KFs can also be treated as several equivalent LTI systems after they reach steady states. Subsequently, for deterministic terms, the analytical expressions of the corresponding output terms can be derived, and for stochastic terms, the theoretical noise filtering performance in the frequency domain can be determined. Finally, by combining the conclusions of the new and traditional methods, complete conclusions similar to (60)–(62) can be obtained as the theoretical KF performance.

Based on this research, it is recommended to use the classic three-state KF [5], [6] when a quadratic frequency drift (third-order) term is involved in the observation. For the second-order LTI systems in this study, according to (51), the steady-state tracking errors caused by a quadratic frequency drift are inferred to increase with time and not to converge. In the future, when the classic three-state KF is used, the KF in the steady state can be equivalent to several third-order LTI systems. The deterministic deviations caused by the quadratic frequency drift can be theoretically derived and inferred to converge to constants. Similarly, if the deterministic deviations are negligible, the three-state KF can effectively track the states.

2) Determining the theoretical performances of KFs when the KF models or $\boldsymbol{Q}$ and R values are not consistent with the actual observations.

In this study, $z_k = A(t_k) + B(t_k) + C(t_k) + D(t_k)$ is used as an example. The core idea of the new method can be applied to any KF input form, such as $z_k = A(t_k) + B(t_k) + C(t_k) + D(t_k) + E(t_k) + F(t_k)\ldots$. The theoretical performance of each term can then be independently determined.

For example, in [8], one deterministic term in the observation is expressed as a moving average (i.e., (1) in [8]), which does not conform to the KF model. Based on this research, the theoretical expressions of its corresponding output terms can be determined and the conclusions can be used to supplement [8].

Other examples of the $\boldsymbol{Q}$ and R values are as follows. In [8], a small value is set to the random run frequency modulation noise component, which is too small to be observed. In [14], the adopted value of $\boldsymbol{Q}$ represents a compromise between detection performance and transient duration. According to this study, different $\boldsymbol{Q}$ and R values correspond to different coefficients (Ks_1 and Ks_2) of the equivalent LTI systems. Soon, the theoretical KF performances in these cases will be able to be qualitatively determined by using the new method.

VIII. CONCLUSION

In this article, we use the classic two-state KF as the research object and propose a new method to determine its theoretical performance regardless of whether the KF model and $\boldsymbol{Q}$ and R values are consistent with actual observations. The core idea is that the steady-state KF can be equivalent to three LTI systems by deriving its transfer functions. These LTI systems are fully determined by the given KF model and $\boldsymbol{Q}$ and R values, which are independent of actual observations. Therefore, for each LTI system, the theoretical performance of each output term corresponding to the input term can be independently confirmed based on the superposition and homogeneity of LTI systems.

The observation model, $z_k = A(t_k) + B(t_k) + C(t_k) + D(t_k)$, is used as an example, and three submethods are proposed for the polynomial, periodic fluctuation, and noise terms in the observation. For the deterministic terms $A(t_k) + B(t_k) + C(t_k)$, the analytical expressions of the corresponding output terms are derived. The deterministic deviations caused by the frequency drift and the amplitude and phase changes of the periodic fluctuation are accurately determined. For the stochastic term $D(t_k)$, the theoretical noise filtering performance is accurately determined in the frequency domain.

By combining the conclusions of the new and traditional methods, the complete conclusions are summarized, as shown in (60)–(62). The theoretical conclusions are

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validated empirically. For example, in the experiment, the periodic fluctuation term $C(t_k)$ in z_k is not included in the KF model (14), but the KF could still effectively track the $\hat{x}_k$ state with less estimation error, as demonstrated by Figs. 6 and 7.

This research indicates that the proposed method can effectively determine the theoretical KF performance when the observation model and $\boldsymbol{Q}$ and R values are not consistent with actual observations. In the future, the method and sub-methods can be applied to GNSS onboard clock estimation, and the core idea of the method can also be utilized to determine the theoretical performance of different KFs in different cases.

APPENDIX

The appendix provides the derivation of (56) in Section IV-C, which describes the relationship between $C_y(t_k)$ and $C_{yint}(t_k)$.

The derivative of $C(t_k)$ is

$$\begin{aligned}\frac{dC(t_k)}{dt} &= 2\pi f_p A_0 \cos\left(2\pi f_p t_k + \varphi_0\right)\\ &= 2\pi f_p A_0 \sin\left(2\pi f_p t_k + \varphi_0 + \frac{\pi}{2}\right).\end{aligned} \tag{A.1}$$

That is, for the derivative of a sine signal, the amplitude is multiplied times $2\pi f_p$, and the phase is 90° ahead of that of the original signal. Conversely, for the integral of a sine signal, the amplitude is divided by $2\pi f_p$, and the phase is delayed by 90°.

This relationship is also approximately true for the following discrete form of integration. As shown in (34), $\hat{y}_{int,k}$ is the integral of $\hat{y}_k$ in the discrete form. The transfer function of the discrete integrator is $Int(z)$ in (36). By substituting the approximations $1 - z^{-1} \approx j2\pi fT$ and $z^{-1} \approx 1$ into $Int(z)$ when $2\pi fT \ll 1$

$$\left|Int\left(f_p\right)\right| \approx \frac{1}{2\pi f_p} \tag{A.2}$$

$$\angle Int\left(f_p\right) \approx -\frac{\pi}{2}. \tag{A.3}$$

By (36), (A.2), and (A.3), obtains (56).

The results in Figs. 2 and 3 and Table I confirm that the approximation error in (56) is very small in the example. As shown in Fig. 2, because both $Hx(f)$ and $Hy_{int}(f)$ are low-pass filters and their outputs, $\{\hat{x}_k\}$ and $\{\hat{y}_{int,k}\}$, are time difference series, it is more intuitive to determine $C_y(t_k)$ indirectly through $C_{yint}(t_k)$ by applying (56).

REFERENCES

[1] P. B. Whibberley, J. A. Davis, and S. L. Shemar, "Local representations of UTC in national laboratories," *Metrologia*, vol. 48, pp. 154–163, 2011.

[2] P. Tavella, "Statistical and mathematical tools for atomic clocks," *Metrologia*, vol. 45, no. 6, pp. S183–S192, 2008.

[3] L. Galleani and P. Tavella, "Time and the Kalman filter," *IEEE Control Syst. Mag.*, vol. 30, no. 2, pp. 44–65, Apr. 2010.

[4] Y. Wu, B. Yang, S. Xiao, and M. Wang, "Atomic clock models and frequency stability analyses," *Geomatics Inf. Sci. Wuhan Univ.*, vol. 44, no. 8, pp. 1226–1232, 2019.

[5] G. W. Huang and Q. Zhang, "Real-time estimation of satellite clock offset using adaptively robust Kalman filter with classified adaptive factors," *GPS Solutions*, vol. 16, no. 4, pp. 531–539, 2012.

[6] G. W. Huang, Q. Zhang, and G. C. Xu, "Real-time clock offset prediction with an improved model," *GPS Solutions*, vol. 18, no. 1, pp. 95–104, 2014.

[7] J. A. Davis, S. L. Shemar, and P. B. Whibberley, "A Kalman filter UTC(k) prediction and steering algorithm," in *Proc. Joint Conf. IEEE Int. Freq. Control Eur. Freq. Time Forum*, 2011, pp. 1–6.

[8] J. Yao, T. E. Parker, N. Ashby, and J. Levine, "Incorporating an optical clock into a time scale," *IEEE Trans. Ultrasonics, Ferroelect., Freq. Control*, vol. 65, no. 1, pp. 127–134, Jan. 2018.

[9] J. Yao et al., "Optical-clock-based time scale," *Phys. Rev. Appl.*, vol. 12, no. 4, pp. 044069–1–044069–10, 2019.

[10] P. Koppang and R. Leland, "Linear quadratic stochastic control of atomic hydrogen masers," *IEEE Trans. Ultrasonics, Ferroelect., Freq. Control*, vol. 46, no. 3, pp. 517–522, May 1999.

[11] S. W. Lee, C. B. Lee, and S. H. Yang, "A new approach for steering UTC(KRIS)," *IEEE Trans. Instrum. Meas.*, vol. 58, no. 4, pp. 1247–1251, Apr. 2009.

[12] K. G. Mishagin, V. A. Lysenko, and S. Y. Medvedev, "A practical approach to optimal control problem for atomic clocks," *IEEE Trans. Ultrasonics, Ferroelect., Freq. Control*, vol. 67, no. 5, pp. 1080–1087, May 2020.

[13] M. Gödel, T. D. Schmidt, and J. Furthner, "Comparison between simulation and hardware realization for different clock steering techniques," *Metrologia*, vol. 56, no. 3, pp. 035001–1–035001–11, 2019.

[14] L. Galleani and P. Tavella, "Detection of atomic clock frequency jumps with the Kalman filter," *IEEE Trans. Ultrasonics, Ferroelect., Freq. Control*, vol. 59, no. 3, pp. 504–509, Mar. 2012.

[15] X. Huang, H. Gong, and G. Ou, "Detection of weak frequency jumps for GNSS onboard clocks," *IEEE Trans. Ultrasonics, Ferroelect., Freq. Control*, vol. 61, no. 5, pp. 747–755, May 2014.

[16] X. M. Huang, H. Gong, X. Zhu, and G. Ou, "Detection of weak frequency anomalies for atomic clocks with a Kalman filter," *Metrologia*, vol. 51, no. 3, pp. 183–190, 2014.

[17] H. Song, S. Dong, W. Wu, M. Jiang, and W. Wang, "Detecting an atomic clock frequency anomaly using an adaptive Kalman filter algorithm," *Metrologia*, vol. 55, no. 3, pp. 350–359, 2018.

[18] L. Galleani and P. Tavella, "On the use of the Kalman filter in timescales," *Metrologia*, vol. 40, no. 3, pp. S326–S334, 2003.

[19] N. Stacey and S. D'amico, "Adaptive and dynamically constrained process noise estimation for orbit determination," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 57, no. 5, pp. 2920–2937, Oct. 2021.

[20] X. Yu and J. Li, "Adaptive Kalman filtering for recursive both additive noise and multiplicative noise," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 58, no. 3, pp. 1634–1649, Jun. 2022.

[21] X. Ning, F. Wang, and J. Fang, "An implicit UKF for satellite stellar refraction navigation system," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 53, no. 3, pp. 1489–1503, Jun. 2017.

[22] S. M. Kay, *Fundamentals of Statistical Signal Processing Volume: Estimation Theory*. London, U.K.: Pearson Educ., 1993.

[23] M. S. Grewal and A. P. Andrews, *Kalman Filtering: Theory and Practice*, 2nd ed. Hoboken, NJ, USA: Wiley, 2001.

[24] K. L. Senior, J. R. Ray, and R. L. Beard, "Characterization of periodic variations in the GPS satellite clocks," *GPS Solutions*, vol. 12, no. 3, pp. 211–225, 2008.

[25] P. Zhou et al., "Periodic variations of Beidou satellite clock offsets derived from multi-satellite orbit determination," *Acta Geodaetica et Cartographica Sinica*, vol. 44, no. 12, pp. 1299–1306, 2015.

[26] I. Sesia, L. Galleani, and P. Tavella, "Application of the dynamic Allan variance for the characterization of space clock behavior," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 47, no. 2, pp. 884–894, Apr. 2011.

[27] D. Matsakis, "Characterizing the diurnal signature in two way satellite time transfer (TWSTT) data with a Kalman filter," in *Proc. 47th Annu. Precise Time Time Interval Syst. Appl. Meeting*, 2016, pp. 132–146.

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[28] L. Xu and L. Zhi-Cai, "A new noise covariance matrix estimation method of Kalman filter for satellite clock errors," *Acta Physica Sinica*, vol. 64, no. 8, 2015, Art. no. 080201.

[29] H.-J. Song et al., "An improved Kalman filter time scale algorithm for atomic clock noise variation," *Acta Physica Sinica*, vol. 69, no. 17, 2020, Art. no. 170201.

[30] Y. Wu, H. Gong, X. Zhu, and G. Ou, "A DPLL method applied to clock steering," *IEEE Trans. Instrum. Meas.*, vol. 65, no. 6, pp. 1331–1342, Jun. 2016.

[31] Y. Wu, H. Gong, X. Zhu, and G. Ou, "A clock steering method: Using a third-order type 3 DPLL equivalent to a Kalman filter with a delay," *Metrologia*, vol. 52, no. 6, pp. 864–877, 2015.

[32] L. Galleani, L. Sacerdote, P. Tavella, and C. Zucca, "A mathematical model for the atomic clock error," *Metrologia*, vol. 40, no. 3, pp. S257–S264, 2003.

[33] L. Galleani, "A tutorial on the two-state model of the atomic clock noise," *Metrologia*, vol. 45, no. 6, pp. S175–S182, 2008.

[34] G. Panfilo and P. Tavella, "Atomic clock prediction based on stochastic differential equations," *Metrologia*, vol. 45, no. 6, pp. S108–S116, 2008.

[35] W. J. Riley, *Handbook of Frequency Stability Analysis*, vol. 1065. Gaithersburg, MD, USA: NIST Special Publication. 2008.

[36] G. F. Franklin et al., *Feedback Control of Dynamic System*, 6th ed. London, U.K.: Pearson Educ., 2010.

[37] F. M. Gardner, *Phaselock Techniques*, 3rd ed. Hoboken, NJ, USA: Wiley, 2005.

[38] A. V. Oppenheim, J. Buck, M. Daniel, A. S. Willsky, S. H. Nawab, and A. Singer, *Signals and Systems*, 2nd ed. London, U.K.: Pearson Educ., 1997.

[39] A. V. Oppenheim, *Discrete-Time Signal Processing*, 2nd ed. London, U.K.: Pearson Educ., 1999.

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